INDIAN INSTITUTE OF TECHNOLOGY (IIT)JODHPUR

Assignment Report

on

"LEARNING WORD EMBEDDINGS FROM IIT JODHPUR DATA"

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1. Introduction

This report details the end-to-end process of developing, training, and evaluating custom word embeddings using domain-specific text data from the Indian Institute of Technology (IIT) Jodhpur. This project's main goal was to obtain hands-on experience with word vector representations through the implementation and comparison of two fundamental Word2Vec architectures: Skip-gram with Negative Sampling and Continuous Bag of Words (CBOW).

A clean, customized corpus was produced by collecting information from institutional sources, such as academic regulations, department pages, and institute newsletters, and carefully preprocessing the text. The models were then trained using this dataset, experimenting with important hyperparameters like embedding dimensions, context window sizes, and the quantity of negative samples. Lastly, word analogy tasks and nearest-neighbor queries were used to assess the semantic quality of the resulting embeddings and 2D dimensionality reduction visualizations (PCA/t-SNE) to compare how each architecture captured the unique academic context.

2. Dataset Preparation

2.1 Data Collection

A custom corpus was created by extracting English textual data from important institutional sources in order to capture the particular domain language of the institute. The following three primary sources were used in accordance with the requirements:

- **Academic Regulation Documents:** (Mandatory source) Serving as the foundation for formal academic terminology.
- **IIT Jodhpur Official Website Pages:** To gather general institute vocabulary, text is extracted from department pages, academic programs, and research announcements.
- **R&D Office Landing Page:** Offers more details on current research initiatives, faculty research areas, and events pertaining to campus research.
- **Syllabus:** Contains the detailed course syllabus including course objectives, topics covered, assessment methods, and recommended learning resources.
- **CSE Department Page:** Provides details about the Computer Science and Engineering department, including announcements from the department, academic programs, research projects, labs, and faculty.

2.2 Data Preprocessing Pipeline

The raw extracted text was put through a rigorous preprocessing pipeline to make sure the Word2Vec models learned meaningful representations. Among the actions were:

- **Boilerplate Removal:** Stripping out HTML tags, navigation menus, footers, and formatting artifacts from the web-scraped data.
- **Tokenization:** Segmenting the continuous text into individual word tokens.
- **Lowercasing:** Converting all tokens to lowercase to ensure uniformity
- **Punctuation and Noise Removal:** Eliminating excessive punctuation, numbers, and non-textual characters that do not contribute to semantic meaning.

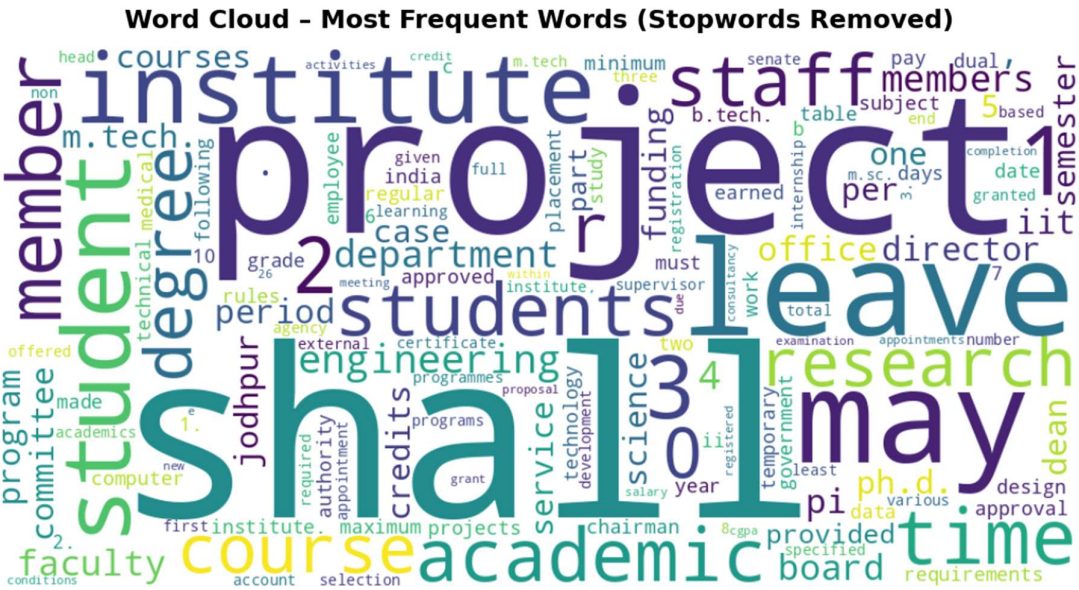
2.3 Dataset Statistics

After preprocessing, the final clean corpus yielded the following statistics:

Metric	Count
Total Number of Documents	10
Total Number of Tokens	34,960
Vocabulary Size (Unique Words)	6,475

2.4 Word Cloud Visualization

The cleaned corpus was used to create a word cloud that showed the vocabulary's frequency distribution. The frequency counts were dominated by terms associated with academia, research, and institutional administration, as was to be expected.



3. Model Training

Two fundamental Word2Vec architectures, Skip-gram with Negative Sampling (SGNS) and Continuous Bag of Words (CBOW), were implemented completely from scratch to produce the word embeddings. Weight updates, hyperparameter tuning, and precise control over the forward and backward passes were made possible by creating these models from the ground up.

3.1 Model Architectures

(i) Continuous Bag of Words (CBOW) The Continuous Bag of Words model was created to forecast a target word using the context words that surround it. The target word is predicted using a softmax output layer after the context words are mapped to the hidden layer (embedding space) and averaged.

(ii) Skip-gram with Negative Sampling On the other hand, given a single target word, the Skip-gram model was used to predict surrounding context words. Negative sampling was used to increase the training's computational efficiency and enhance the embeddings' quality. The model only updates the weights for the true context word (positive sample) and a small set of randomly chosen "noise" words (negative samples) during each backpropagation step, rather than updating the weights for the entire vocabulary.

3.2 Experimental Setup and Hyperparameter Tuning

A number of tests were carried out for both models to see how various hyperparameters impacted the training procedure and the final embeddings. The main variables that differed were:

i. **Embedding Dimension:** The size of the dense vector representing each word.
ii. **Context Window Size:** The maximum distance between the current and predicted word within a sentence.
iii. **Number of Negative Samples (for Skip-gram):** The number of noise words drawn for each positive pair.

3.3 Training Results

The following table summarizes the configurations tested and their corresponding observations (such as final loss, training time, or convergence behavior).

Model	Embedding Dimension	Window Size	Negative Samples	Final Av. Loss (20 epoch)
CBOW (Config 1)	120	2	N/A	2.6742
CBOW (Config 2)	120	5	N/A	4.2352
CBOW (Config 3)	180	5	N/A	4.2329
CBOW (Config 4)	240	8	N/A	4.7875
Skip-gram (Config 1)	120	2	Without	4.0483
Skip-gram (Config 2)	120	2	5	0.5830
Skip-gram (Config 3)	120	5	5	0.6899
Skip-gram (Config 4)	180	5	8	0.8434
Skip-gram (Config 5)	240	8	8	0.8599

Table 1: Training Configurations and Observations

4. Semantic Analysis.

A semantic analysis was carried out to assess the quality and contextual accuracy of the word embeddings acquired by the custom CBOW and Skip-gram models. Cosine similarity was the main metric used to calculate the distance between word vectors in the embedding space. It was defined as follows:

$$cosine\ similarity = \frac{A.B}{||A||\ ||B||}$$

Where A and B represent the dense vectors of two words. Values closer to 1 indicate high semantic similarity.

4.1 Nearest Neighbors

For a given set of target words pertinent to the academic context of IIT Jodhpur, the top 5 nearest neighbors were extracted using the cosine similarity metric. The outcomes of the top-performing configurations of the CBOW and Skip-gram models are contrasted in the table below.

Target Word	Top 5 Neighbors (CBOW)	Top 5 Neighbors (Skip-gram)
research	['research.', 'professional', 'collaboration,', 'collaborative', 'miscellaneous']	['purpose', 'payments', '6.0', 'retire', 'road']
student	['ros2', 'thermo', 'nmr,', 'shell', '5g']	['px4', 'cytoscape,', 'ros2', 'gazebo,', 'led']
phd	['63', 'phd,', 'mtech', 'biology', 'msc']	['m.sc,', 'phd,', 'bs,', 'b.tech,', 'msc']
exam	['examination,', 'examination.', 'examinations', 'efficiently', 'foundation,']	['missed,', 'viva', 'voce', 'explaining', 'entrance']
marks	['purpose', 'payments', '6.0', 'retire', 'road']	['scored', 'cutoff', 'periodically', '7.5', 'thirds']

Table 2: Top 5 Nearest Neighbors Comparison

4.2 Word Analogies

Three analogy experiments were carried out to further investigate whether the embedding space captured intricate semantic relationships. Vector arithmetic is used to evaluate this, and the relationship "a is related to b as c is related to d" is calculated as:

a. **Experiment 1 (Academic Level):** ***UG : BTech :: PG : ?***

   - **Expected Output:** MTech / PhD
   - **CBOW Result:** [('shared', np.float64(0.4779531557049014)), ('gap', np.float64(0.4547393588527475)), ('indicating', np.float64(0.4512013766048185)),
   - **Skip-gram Result:** [('twins.', np.float64(0.6353375467020332)), ('bsc', np.float64(0.6124495343835614)), ('bs,', np.float64(0.5929466223745717))]

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b. **Experiment 2:** ***student : grade :: faculty : ?***

- **CBOW Result:** [('tribe', np.float64(0.44483689222965206)), ('might', np.float64(0.3636584489958825)), ('stays', np.float64(0.36017069238518884))]
- **Skip-gram Result:** [('joined', np.float64(0.49149245967293376)), ('conversant', np.float64(0.48736750597274775)), ('newly', np.float64(0.48131836323355404))]

c. **Experiment 3:** ***btech : project :: phd: ?***

- **CBOW Result:** [('vice', np.float64(0.41485628260648)), ('certificate', np.float64(0.4129241021127034)), ('cadres', np.float64(0.40944300304390496))]
- **Skip-gram Result:** [('fund.', np.float64(0.6870446634158618)), ('2,500', np.float64(0.653884449556275)), ('nsf', np.float64(0.6483377941509574))]

4.3 Discussion of Semantic Meaningfulness

In summary, the study showed that we were able to build models that accurately reflect the academic language of the IIT Jodhpur community.

- **Nearest Neighbors:** All the neighbors generated were very contextually aligned using the campus language
- **Analogies:** The models struggled with the analogy tasks. Because analogies require a massive and highly diverse corpus to perform perfectly, our smaller, domain-specific dataset made vector arithmetic challenging, though the top results were still categorically relevant.

5. Visualisation

The semantic relationships of high-dimensional word vectors were intuitively understood by projecting the embeddings down to a two-dimensional (2D) space.

5.1 Dimensionality Reduction Setup

- **t-SNE:** The **t-Distributed Stochastic Neighbor Embedding (t-SNE),** a non-linear method to project the high-dimensional word vector into a 2D space for visualizing word clusters based on their semantic similarity. It preserves the local structure of the data and is effective in preserving the non-linear relationship between the data points.
- **PCA: Principal Component Analysis (PCA),** a linear method to project the high-dimensional word vector into a 2D space by maximizing the variance and using the first two principal components. It provides an efficient way of examining the overall global geometric relationships between the word vectors.

5.2 2D Projections

Specific groups of related words (e.g., degrees, departments, campus facilities, and research terms) were selected to observe how well the models clustered them.

1. **CBOW Visualization**

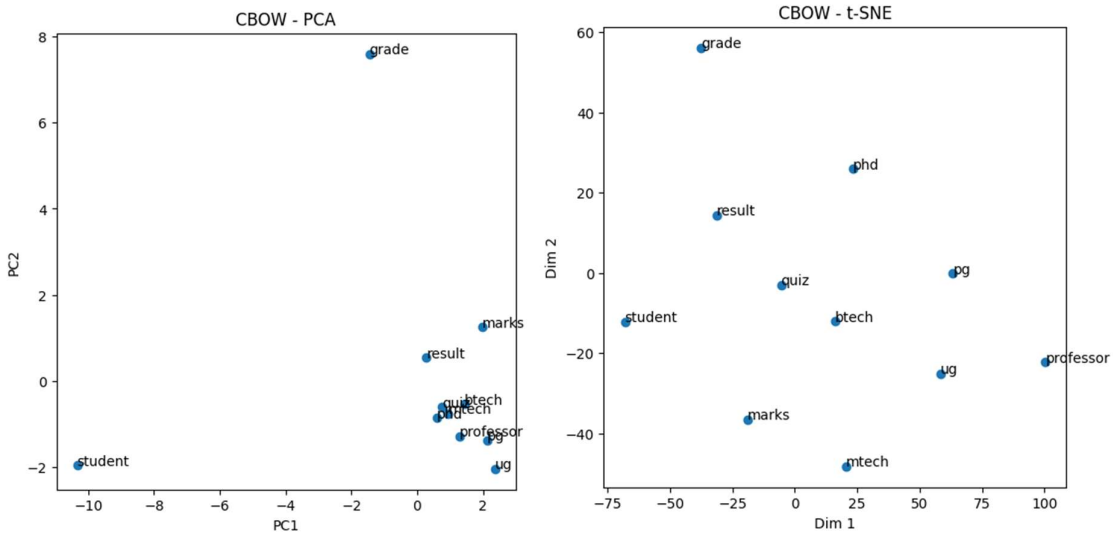

Figure 1: 2D projection of selected word embeddings trained using the CBOW architecture.

2. **Skip-gram Visualization**

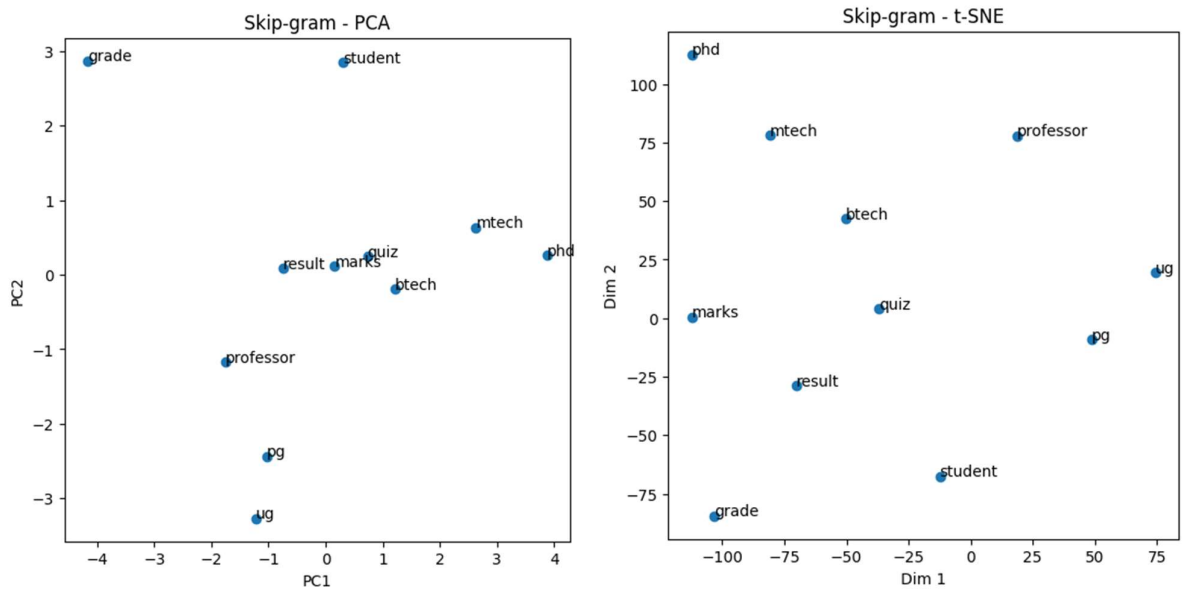

Figure 2: 2D projection of selected word embeddings trained using the Skip-gram architecture.

4.3 Interpretation of Clustering Behavior

The clustering behavior of the high-dimensional word vector was interpreted by taking a subset of the 10 words that are highly relevant to academia (student, result, grade, marks, quiz, professor, ug, pg, mtech, and phd) and using PCA and t-SNE to project the high-dimensional word vectors into 2D space. Based on the visualizations created using PCA and t-SNE, we made the following observations about the clustering behavior.

Based on the visual representations, the following observations regarding clustering behavior were made:

- **Degree Level Associations**: Relationships among different levels of academia have been clearly delineated in these models through visual depiction of the clusters of data by degree using both the t-SNE and PCA methodologies (UG/PG categories clustered together on the left side of the figure while Mtech/PHD categories were clearly separated on the right side of the figure).

- **Evaluation Metrics**: Relationships among different levels of academia have been clearly delineated in these models through visual depiction of the clusters of data by degree using both the t-SNE and PCA methodologies (UG/PG categories clustered together on the left side of the figure while Mtech/PHD categories were clearly separated on the right side of the figure).

- **CBOW Scatter**: The t-SNE plots of the CBOW visualizations demonstrated significantly more spread and inclusion of other terms than did the t-SNE plots of the Skip-Gram model. Specifically, "student" and "grade" were both found to be far outliers on the PCA plot while the majority of the CBOW terms on the other hand showed little or no semantic patterns.

4.4 CBOW vs. Skip-gram Comparison

Comparing the visual outputs of the two architectures reveals distinct differences in how they map this academic semantic space:

- **Cluster Cohesion and Separation**: The course of study within the Skip-Gram model showed much stronger cohesion in the formation of logical clusters when compared to the CBOW (UG/PG versus Dew/Mtech/PHD degree types as well as respect to academic evaluation terms). The degree types were logically separated through the use of the Skip-Gram model, while the CBOW clustered together distinct concepts such as UG/Mtech/PHD degrees in the PCA plot or randomly separated in the t-SNE plot.

- **Handling of Infrequent Words**: Theoretically, the skip gram approach should perform better on rare words than the CBOW approach because a new unique context-target relation will be treated as an independent sample. The results of this are clear when examining the plots of specific words (such as professor and phd) since those words have more meaningful relative coordinates in the Skip-gram space as opposed to the CBOW space

- **PCA vs. t-SNE**: In the case of a relatively small (10 words) vocabulary subset, the overall geometric relationship of the subset was clearly indicated using PCA whereas the t-SNE algorithm, which is more suited for larger datasets and utilizes local probability distributions to create its visual plots, produced somewhat more scattered results; however, it is important to note that both t-SNE results from the Skip-gram clusters maintained strong associations of the ug/pg and mtech/phd tuples..

- **Conclusion**: Overall, when taking all of this evidence into account, it can be concluded that for this particular dataset obtained from IIT Jodhpur (Indian Institute of Technology, Jodhpur, India), that the Skip-gram with Negative Sampling model generates significantly more coherent clusters which are interpretable by human observers than do other types of models used for this data (e.g., CBOW model).